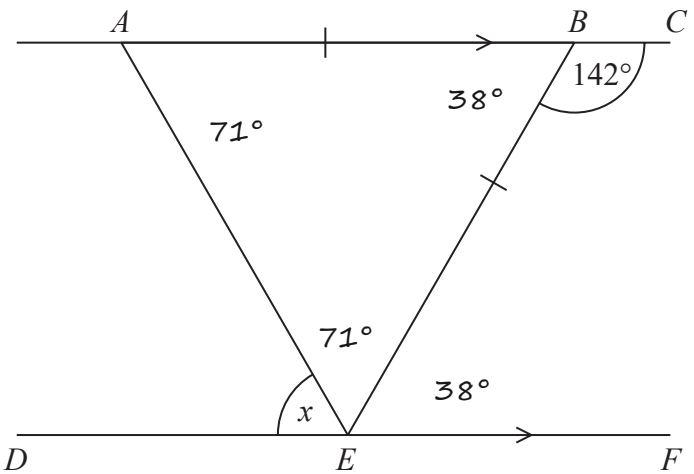
19



ABC and *DEF* are parallel straight lines.
ABE is an isosceles triangle with *AB* = *BE*.
Angle *CBE* = 142°

Work out the size of angle *x*.
Give a reason for each stage in your working.

Angle ABE = 180° – 142° = 38°.

Angles on the straight line ABC add to180°.

1 mark
Straight line, equal base angles; triangle

AB = BE, so base angles BAE and AEB are equal.

Angle AEB = (180° – 38°) ÷ 2 = 71°.

1 mark
Complete angle chain

Angles in triangle ABE add to180°.

x = angle BAE = 71°.

1 mark
Use parallel lines

Alternate angles are equal: ABC is parallel to DEF.

1 mark
Alternate angles

1 mark
71°

°

(Total for Question 19 is 5 marks)